University of Florida
College of Public Health and Health Professions
PHC 6740: Public Health Methods 1: Quantitative Foundations (3 credits)
Semester: Spring 2026
Delivery Format: Online asynchronous

Instructor Name: Jonathan Fischer, PhD
Class Locations: Canvas
Office Location: CTRB 5223
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Office Hours: Mondays and Tuesdays 3-4 PM; others by appointment
Preferred Course Communications:

- For specific questions or to discuss issues of a personal nature, use Canvas messages.
- For general questions, use the Canvas discussion board.
- If you send me an email, please double-check the email address. There is an individual with a very similar email address, and sometimes students mistakenly send their emails to him. I generally try to reply within 1-2 business days.

Prerequisites: None
Corequisites: PHC 6741 Public Health Methods 1 Lab: Applying Statistical Methods

I. PURPOSE & OUTCOME

Course Overview

This is the first of two courses that focus on public health/global health research and practice methods. This course, taken in conjunction with a lab, focuses on principles of epidemiology and biostatistics, emphasizing application of epidemiological methods, quantitative data collection, and quantitative data analysis and interpretation.

Relation to Program Outcomes

This course is associated with the following MPH core competencies as outlined by the Council on Education for Public Health (CEPH, 2024, Section D2):

- Core 1: Apply epidemiological methods to the breadth of settings and situations in public health practice.
- Core 2: Select quantitative and qualitative data collection methods appropriate for a given public health context.
- Core 3: Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming and software, as appropriate.
- Core 4: Interpret results of data analysis for public health research, policy, or practice.

Course Objectives/Outcomes

1. Identify basic ethical and legal principles pertaining to the collection, maintenance, use, and dissemination of data.

2. List, define, and calculate basic epidemiology measures of disease frequency and comparisons of disease risk between populations.
3. Select a study design, data collection methods, and data analysis to appropriately address a specific research question and/or hypothesis.
4. Differentiate among different sampling methods and discuss their strengths and limitations within the context of study design.
5. Evaluate the quality of data for purposes of data analysis, interpretation, and reporting.
6. Use biostatistics, computer-based programming, and/or software for data input, management, summarization, visualization, and analysis.
7. Assess the validity of data analysis methods.
8. Interpret results of quantitative data analysis for public health research, policy, or practice.
9. Discuss the science of primary, secondary, and tertiary prevention in population health, including health promotion, screening.

Instructional methods

1. Readings: Required readings are assigned for each module. Students are responsible for this material and will complete weekly quizzes based on that week's readings. These are indicated in the schedule below.

2. Lectures: Recorded lecture presentations will be provided in Canvas and are to be watched prior to class sessions.

3. Quizzes: Each week, a short quiz associated with the reading and lecture recording is to be taken prior to class. Students will have up to 3 attempts on each quiz, with the best score counting towards the course grade.

4. Homework: Each week, a homework assignment related to the current module will be assigned. In each week's problem session, we will either work on the homework assignment directly or work on a similar document. These assignments can be completed in groups or individually (see assignments section below) and, if necessary, are to be finished after class. Of submitted homework assignments, the lowest score will be dropped.

5. Exams: A midterm and final exam will be given and administered with proctoring software.

II. DESCRIPTION OF COURSE CONTENT, FORMAT, & ACCESS

Topical Outline/Course Schedule

Week of	Topic(s)	Assignments	Readings
1/12	Introduction to Biostatistics & Epidemiology • Sources of data • Elements of descriptive epi	Q0, Q1, HW1	BOLT Introduction, EEPH Chapters 1, 4 Suggested: Merrill Ch 1, Introduction
1/19	Variables and data, random samples	Q2, HW2	BOLT Sections 1.1-1.3, 2.1 Suggested: Merrill Ch 2, Data & Descriptive Measures

Week of	Topic(s)	Assignments	Readings
1/26	Probability, independence and disjointness	Q3, HW3	BOLT Section 3.1 Suggested: Merrill Ch 6, Probability
2/2	Random variables and distributions	Q4, HW4	BOLT Sections 3.2 – 3.3 Suggested: Merrill Ch 7, Random Variables & Probability Distributions
2/9	Point and interval estimation	Q5, HW5	BOLT Sections 4.1.1 – 4.1.4
2/16	Hypothesis testing	Q6, HW6	BOLT Sections 4.1.5-4.1.7 Suggested: Merrill Ch 8, Estimation & Hypothesis Testing
2/23	Measures of disease frequency • Populations in Epidemiology • Ratio, Proportion, Rate • Prevalence and Incidence • Case definitions • Rates: Mortality rate, survival rate, attack rate, case fatality rate	Q7, HW7	EEPH Chapter 2
3/2	Comparing measures • Organize a 2x2 table • Comparison of disease frequency • Measures of association • Attributable risk • Attributable proportion • Relative risk • Standardization	Q8, HW8	EEPH Chapter 3 Suggested: Merrill Ch 3, Standardizing Rates Suggested: Merrill Ch 11, Statistical Measures of Association Among Variables
3/9	Exam 1 (Modules 1 - 8)		

Week of	Topic(s)	Assignments	Readings
3/23	Screening in Public Health Practice • Natural History of Disease • Screening methods • Primary, secondary, & tertiary prevention • Characteristics that make a disease appropriate for screening • Sensitivity, specificity, and positive and negative predictor values of a screening test • Potential for bias in screening programs	Q9, HW9	EEPH Chapter 16
3/30	Experimental studies • Conducting Epidemiologic studies • experimental and observational studies • Types of Experimental study designs • Key features of conducting randomized controlled trials	Q10, HW10	EEPH Chapter 6 Suggested: Merrill Ch 12, Experimental Studies
4/6	Cohort studies • Conducting Cohort study designs • Types of Cohort studies • Key features of Cohort studies • Strengths & limitations of cohort study designs	Q11, HW11	EEPH Chapter 8 Suggested: Merrill Ch 10, Study Designs (p. 207-209)
4/13	Case-Control studies • Case-control study designs • Measures of association: Odds ratios • Ecologic study design • Comparison of study design	Q12, HW12	EEPH Chapter 9 Suggested: Merrill Ch 10, Study Designs (p. 201-207)

Week of	Topic(s)	Assignments	Readings
4/20	Bias, confounding, and causation • Bias • Confounding • Causation • Methods for controlling bias and confounding • Advanced data types	Q13, HW13	EEPH Chapters 10, 11, and 13 Suggested: Merrill Ch 10, Study Designs (p. 209-214) Suggested: Merrill Ch 14, Cause and Effect Provided Notes
4/27	Exam 2 (Modules 9 - 13)		

Course Materials & Technology
This course will use the Canvas CMS. If you experience technical difficulties, please contact the UF Help Desk (learning-support@ufl.edu; 352-392-HELP – select option 2).

Required Materials
(BOLT) – The Biostatistics Open Learning Textbook is a Canvas shell containing readings and tutorials developed by the UF Department of Biostatistics.

Required Textbook:
(EEPH) - Essentials of Epidemiology in Public Health, 4th Edition by Ann Aschengrau, ScD, and George Seage III, DSc. Copyright 2020, ISBN:978-1284128352.

Optional textbook and reference materials:
- Merrill, R. M. (2012): Fundamentals of Epidemiology & Biostatistics. Jones & Bartlett Learning.
- Daniel, W.D. (2013): Biostatistics: A Foundation for Analysis in the Health Sciences. 10th Edition, Wiley.
- Agresti, A. (2013): The Art and Science of Learning from Data. 4th Edition, Pearson.

Additional Academic Resources (see Section V for additional student services)
- Career Connections Center: Reitz Union 1300, 352-392-1601. Career assistance and counseling services.
- Library Support: Receive assistance in using the libraries or finding resources.
- Teaching Center: General study skills and tutoring
  1317 Turlington Hall, 352-392-2010; or, to make a tutoring appointment: 352-392-6420.
- Writing Studio*:* Help with brainstorming, formatting, and writing papers. 2215 Turlington Hall*,* 352-846-1138.
- Student Complaints On-Campus: Visit the Student Honor Code & Student Conduct Code page for more info.
  On-Line Students Complaints: View the Distance Learning Student Complaint Process.

III. Academic Requirements & Grading

Assignment type	Number	Points
Quizzes	14	14 x 10 = 140

Homework	12	12 x 30 = 360
Exams	2	2 x 250 = 500
Total		**1000**

Grading Scale

Percent Earned	[93, 100]	[90, 93)	[87,90)	[83, 87)	[80,83)	[77,80)	[73,77)	[70,73)	[67, 70)	[63,67)	[60, 63)	< 60
Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
Grade Points	4.0	3.67	3.33	3.0	2.67	2.33	2.0	1.67	1.33	1.0	.67	0

Please be aware that grades of C- (or below) are not acceptable for graduate students. Graduate students' GPA must be at least 3.0 in all graduate courses (≥5000 level). A grade of C will count toward the graduate degree only if there are sufficient credits in graduate courses earned with a B+ or higher. Information on current UF grading policies can be found at: https://gradcatalog.ufl.edu/graduate/regulations/#Grades

IV. CLASS POLICIES

Academic & Personal Integrity: I expect and assume that you will be honest with me in all aspects of your conduct regarding our course. In return, I will do the same with you. By formally registering for coursework at the University of Florida, you are bound by the Honor Pledge which states:

> *"We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity by abiding by the Honor Code."* On all work submitted for credit at the University of Florida, the following pledge is either required or implied: *"On my honor, I have neither given nor received unauthorized aid in doing this assignment."*

The Honor Code specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. Violations of the Honor Code will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action.

Attendance Policy: Attendance is required. Students must notify the instructor via email about upcoming absences, late arrival, or if they will need to leave class early as soon as they become aware of potential conflicts. If you miss class without notification or you provide an unacceptable excuse, you will be counted as absent and will not be allowed to make-up assignments missed. Excused absences must be consistent with university policies in the Graduate Catalog.

Course Evaluations: I value your professional and respectful feedback on the quality of instruction in this course. Please complete the evaluation for this course via the Canvas 'GatorEvals' tab or through https://ufl.bluera.com/ufl/. You will be notified when the evaluation period opens. UF provides guidance on how to give feedback in a professional and respectful manner. You can also view public summaries of course and instructor evaluation results.

Policy Related to Make up Exams or Other Work: Please notify me as soon as possible if unanticipated circumstances arise which interfere with your ability to complete an assignment on-time. Late work without prior notification and approval (unless there is an emergency) will receive a zero. Late work with prior approval (or due to an emergency) will be eligible for full credit. Any requests for make-ups due to technical issues must be accompanied by the ticket number received from e-learning support when the problem was reported to them. The ticket number will document the time and date of the problem. You must e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up.

Recording Within the Course: Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

A "class lecture" is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session.

Publication without permission of the instructor is prohibited. To "publish" means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third-party note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.

Title IX: University of Florida has zero tolerance for sexual discrimination, harassment, assault/battery, dating violence, domestic violence, or stalking. Students are encouraged to report any experienced or witnessed occurrences to law enforcement and/or one of UF's Title IX Coordinators. Students can report incidents or learn more about their rights and options here. Or contact Student Conduct and Conflict Resolution at 202 Peabody Hall, 352-392-1261.

V. STUDENT EXPECTATIONS, ROLES, AND OPPORTUNITIES FOR INPUT

Expectations Regarding Course Behavior:

- Complete all work as assigned.
- Take responsibility for the quality of your learning experience.
- Build on one another's comments/ideas; seek to understand others' perspectives.
- Respectfully provide and receive specific, solution-oriented feedback.
- Communicate with your instructor and with your group members.
- Only use electronic devices for course purposes during class meetings.

- Please also refer to academic and personal integrity section under Class Policies.

Netiquette, Communication Courtesy: All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats. I expect that students will show respect to their peers and instructor in all online communications. I will not tolerate improper language and disparaging comments; these actions will result in disciplinary action. Review this resource for information on the expected behavior of students when communicating with peers and instructors online.

VI. SUPPORT SERVICES

Accommodations for Students with Disabilities or Different Abilities: The Americans with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protections for persons with disabilities. Among other things, it requires that all students with documented disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you have a disability (or different-ability) that affects your learning, please reach out to the Disabilities Resource Center (DRC). And then share your accommodation letter with your instructor as quickly as possible to ensure you have access for the full semester.

If you did not register formally, but you know you have different learning, behavioral, or other needs that may affect your performance in the course, tell me and I will help you.

Counseling and Student Health

Students sometimes experience stress from academic expectations and/or personal and interpersonal issues that may interfere with their academic performance. If you find yourself facing issues that have the potential to, or are already, negatively affecting your coursework, please talk with an instructor and/or seek help through University resources available to you.

- Counseling and Wellness Center: Individual counseling, group counseling, and online resources are available to UF students at no charge. Also psychological assessment, intervention, and assistance for math and test anxiety. Visit the website or call 352-392-1575. If you are having a crisis, you can call anytime and ank to speak to the counselor on call.
- GatorWell Health Promotion services: GatorWell provides health-related resources, information, and individual services to students. Recommended services: Wellness Coaching for Academic Success (virtual appointments available).
- The Student Health Care Center , 352-392-0627, at Shands is a satellite clinic of the main Student Health Care Center located on Fletcher Drive on campus. Student Health at Shands offers a variety of clinical services. The clinic is located on the second floor of the Dental Tower in the Health Science Center.
- UMatter, We Care: Available for students who are experiencing personal life disruptions that may affect their academics. UMatter can help you identify resources and communicate with instructors on your behalf. 352-294-CARE (2273), umatter@ufl.edu
- University Police Department: Visit their website or call 352-392-1111 (or 9-1-1 for emergencies).
- Alachua County Crisis Center: Visit the website or call the hotline - 352-264-6789
- Meridian Behavioral Healthcare, 352-374-5600
- **UF Health Shands Emergency Room / Trauma Center:** For immediate medical care call 352-733-0111 or go to the emergency room at 1515 SW Archer Road,

Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center website.

Do not wait until you reach a crisis to come in and talk with us. We have helped many students through stressful situations impacting their academic performance. You are not alone so do not be afraid to ask for assistance.

Inclusive Learning Environment: Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act."